

Tutorial and Assignment

- 1 Draw the Network Security Model and explain it.
- 2 Explain DES construction in detail with a neat diagram
- 3 Explain Playfair Cipher and its rules for the following example: Ex: Keyword: "Computer"
Plaintext: "parrot"
- 4 Using Hill Cipher to encipher and decipher the message "Hi". Use the key (03 02
05 07)
- 5 Explain Feistel encryption and decryption algorithm with neat diagram
- 6 Explain Caesar Cipher with example
- 7 Explain the public key cryptosystem with a neat diagram.
- 8 List the difference between Conventional and public key encryption
- 9 Let $q=353$ and $g=3$. $X_a=97$, $X_b=233$. Use Diffie Hellman Key exchange algorithm to find Y_a , Y_b and Secret key K .
- 10 Explain the Description of RSA Algorithm. Assume 2 prime no as $p=3$ and $q=11$ encryption key $e=3$ apply RSA algorithm & find cipher text for the plaintext $m=8$ & decrypt the cipher text to obtain plain text.
- 11 What is the one-time pad technique? List the advantages and disadvantages.
- 12 Explain Playfair Cipher and its rules for the following example:
Ex: Keyword: "Computer". Plaintext: "parrot"
- 13 Let $q=353$ and $g=3$. $X_a=97$, $X_b=233$. Use Diffie Hellman Key exchange algorithm to find Y_a , Y_b and Secret key K .
- 14 Explain elliptic curve cryptography algorithm
- 15 With a neat diagram, explain the Man-in-middle Attack.